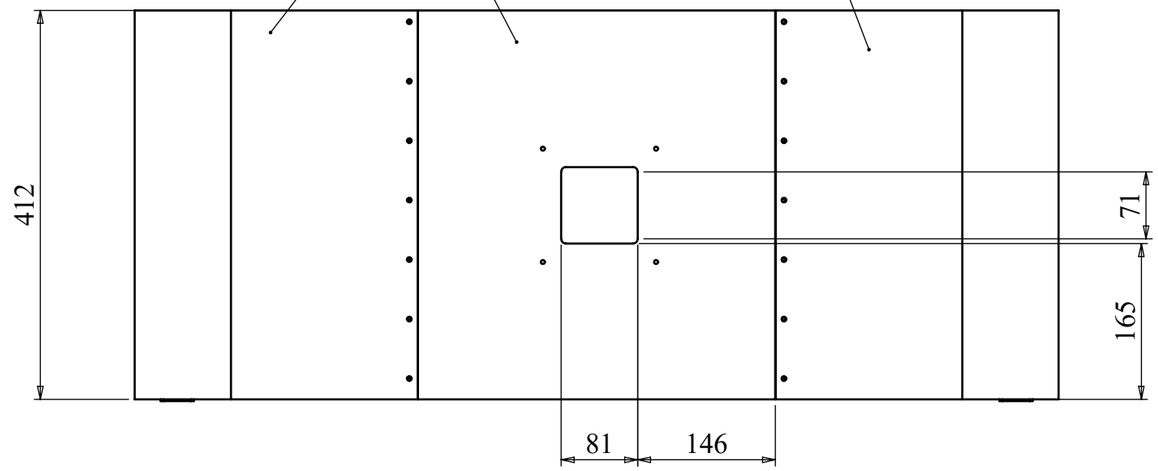
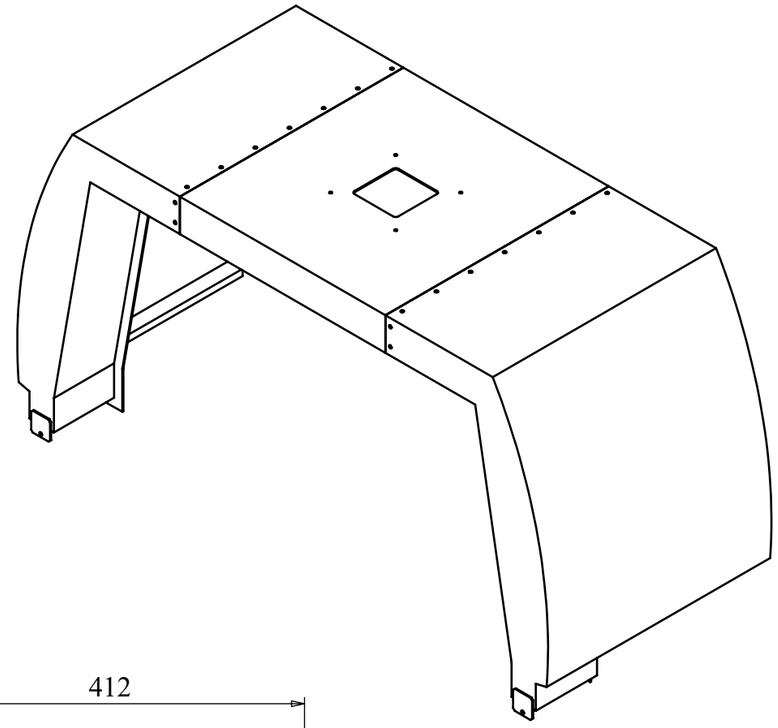


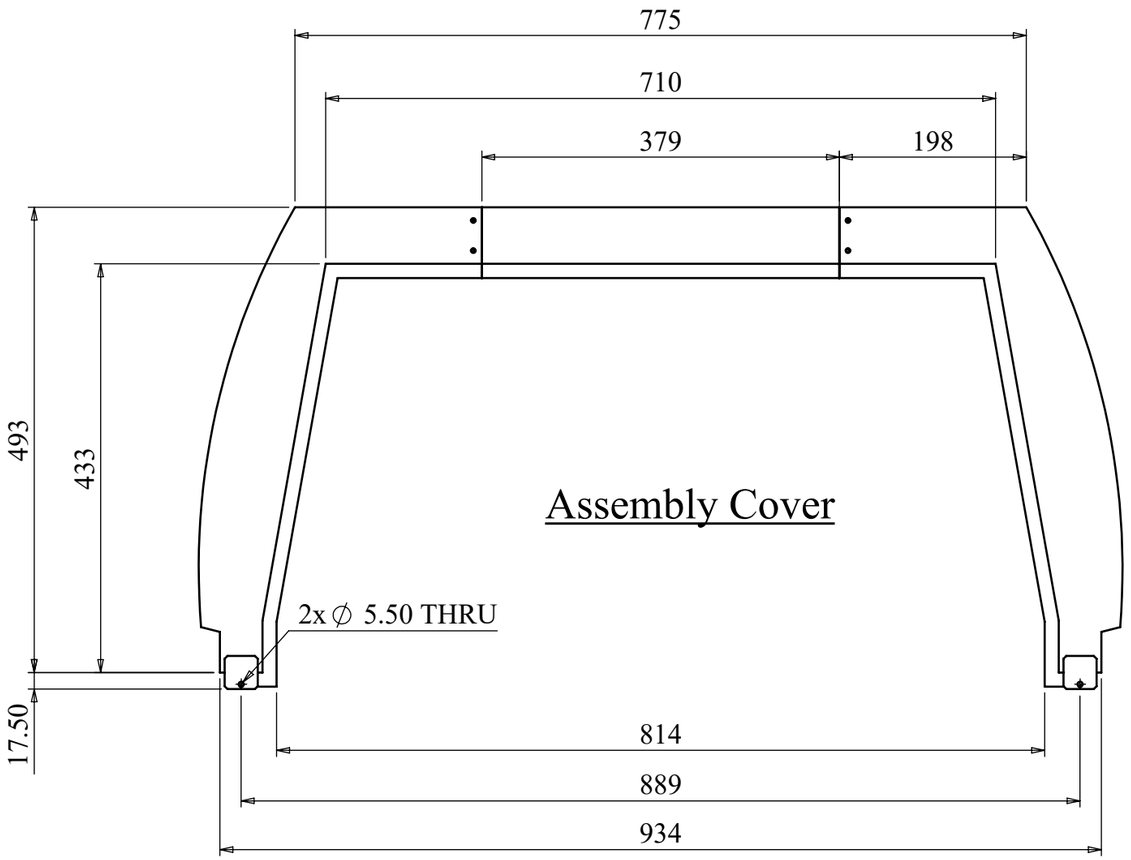
Right Cover
(Detail See Drawing)

Center Cover
(Detail See Drawing)

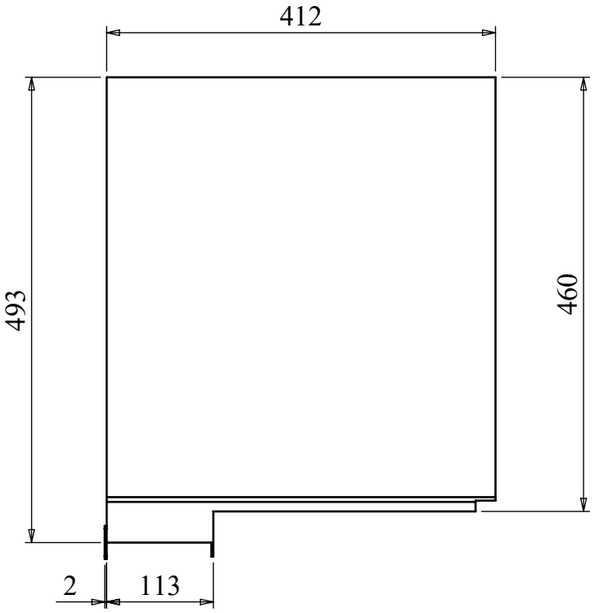
Lift Cover
(Detail See Drawing)







Assembly Cover



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Note

1. Unless otherwise specified all bending edges and corners is R1.0~R5.0
2. Accept welding
3. Paint the primer before the actual color.
4. Rework screw after actual color.
5. Baked Color ***(Ref.) Plate Master : DV-Minebea)
6. Vendor Assembly

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED C0.2 MAX or R0.2 MAX	
GENERAL TOLERANCE	
JIS B 0405 MACHINING WORKING	
LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S / 6.3S / 25S / 100S / UNCUT

M / C: Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL: SPCC

FINISH: Powder Coating

HEAT TREATMENT: -

REMARK

3RD ANGLE PROJECTION

SCALE 1:6.5 | SHEET 1 / 4 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

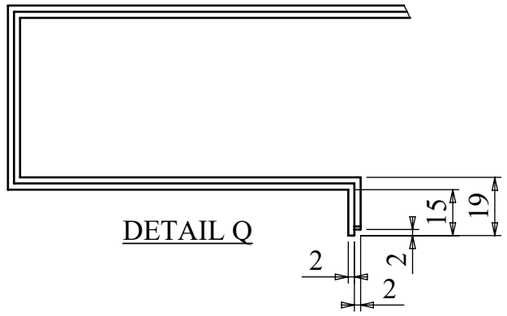
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

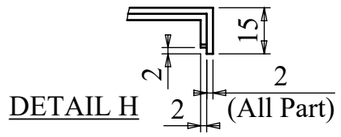
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

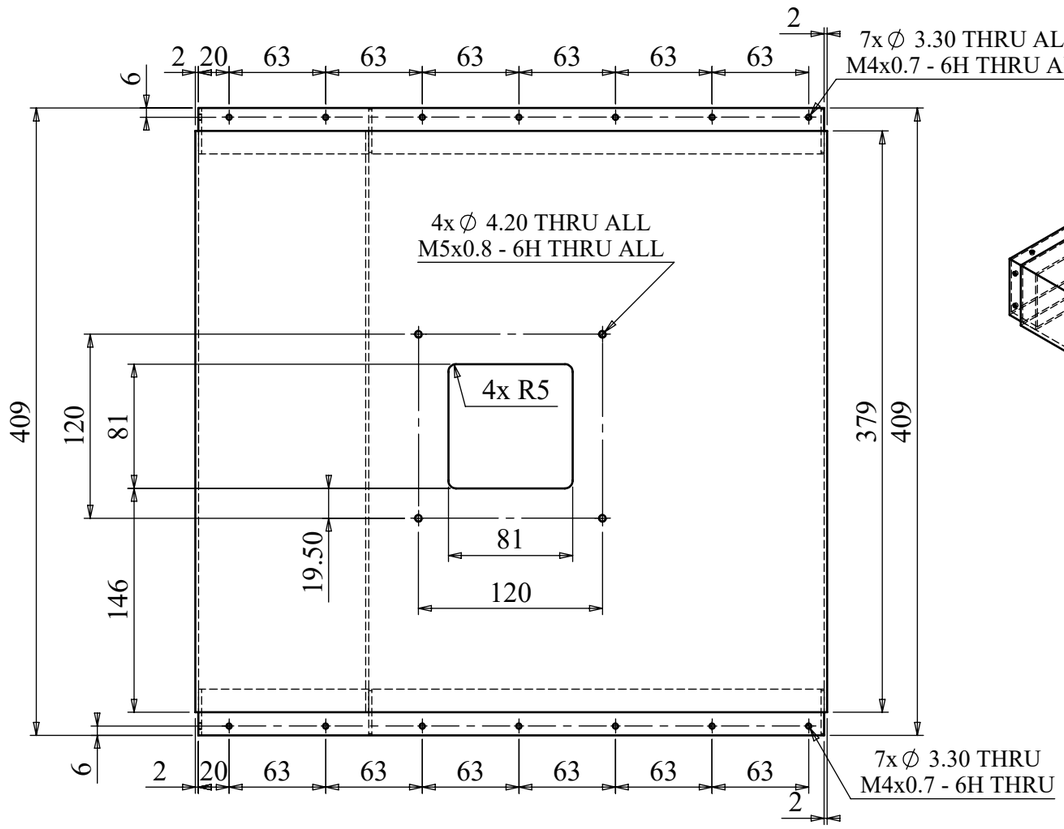
TITLE: Cover Door

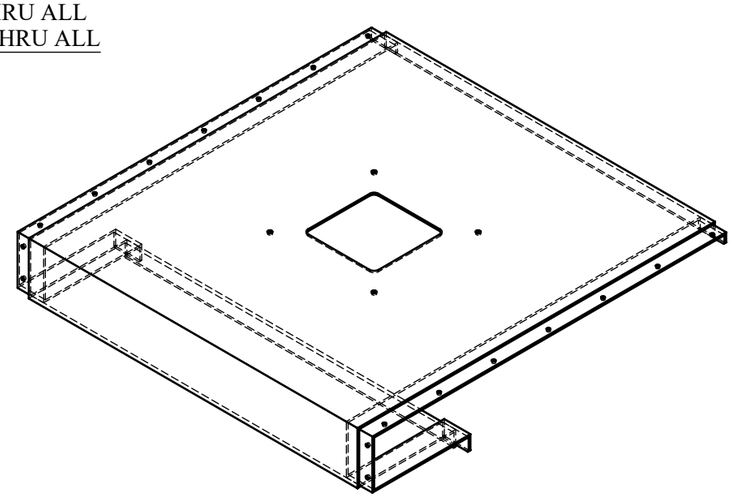
DRAWING NUMBER: DVGM-ALW-MC-0010

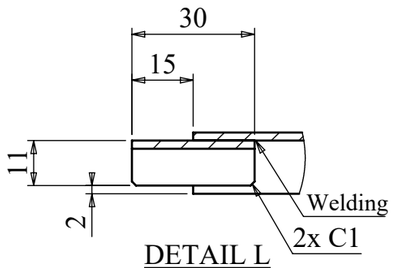
1 REV.



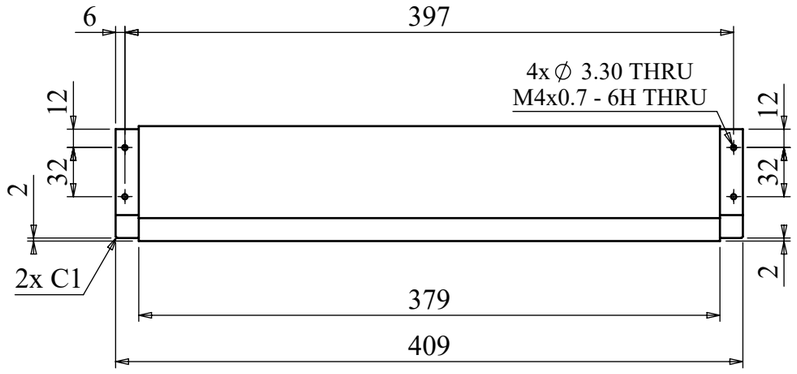


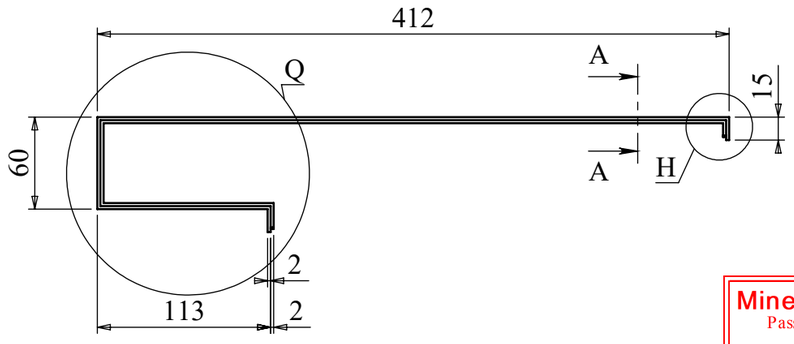


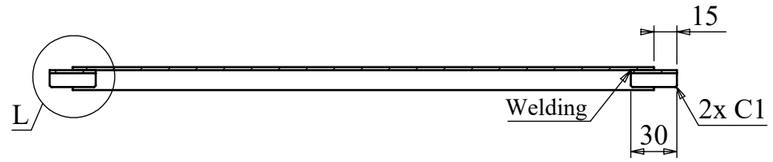




DETAIL L







SECTION A-A

Center Cover

Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Note

1. Unless otherwise specified all bending edges and corners is R1.0~R5.0
2. Accept welding
3. Paint the primer before the actual color.
4. Rework screw after actual color.
5. Baked Color ***(Ref.) Plate Master : DV-Minebea)

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED C0.2 MAX or R0.2 MAX	
GENERAL TOLERANCE	
JIS B 0405 MACHINING WORKING	
LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C: Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL: SPCC

FINISH: Powder Coating

HEAT TREATMENT: -

REMARK

3RD ANGLE PROJECTION

SCALE 1:4 | SHEET 2 / 4 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

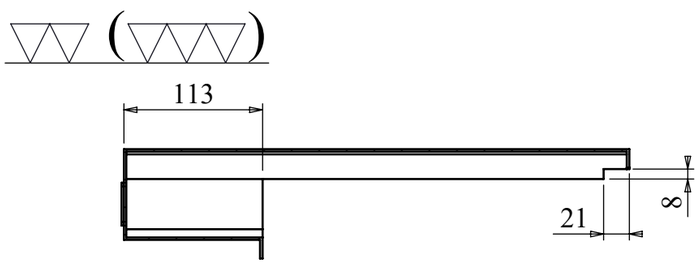
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

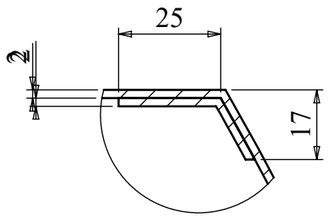
TITLE: Cover Door

DRAWING NUMBER: DVGM-ALW-MC-0010

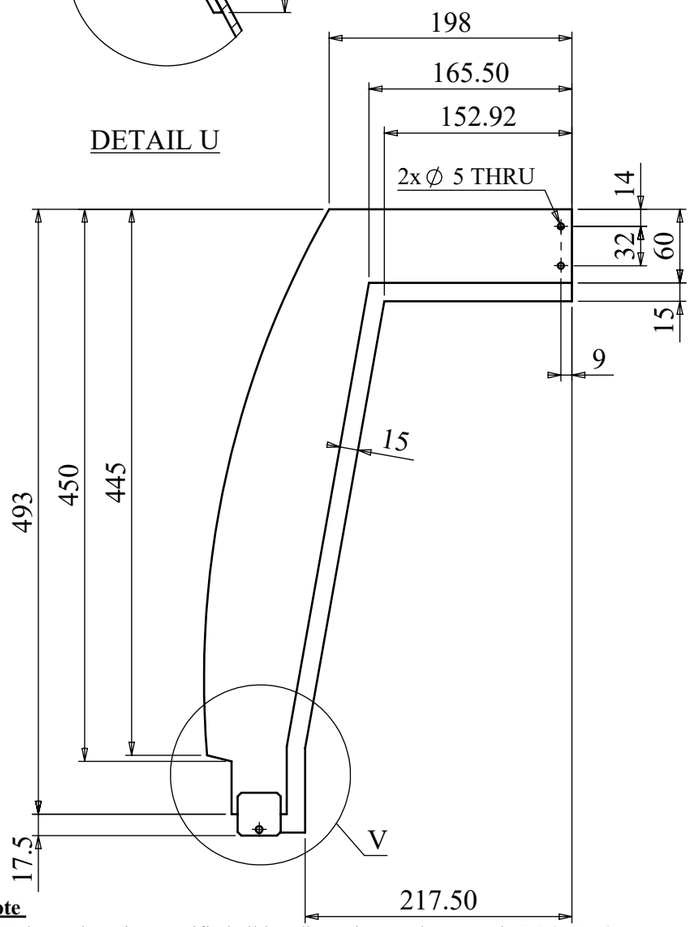
1 REV.

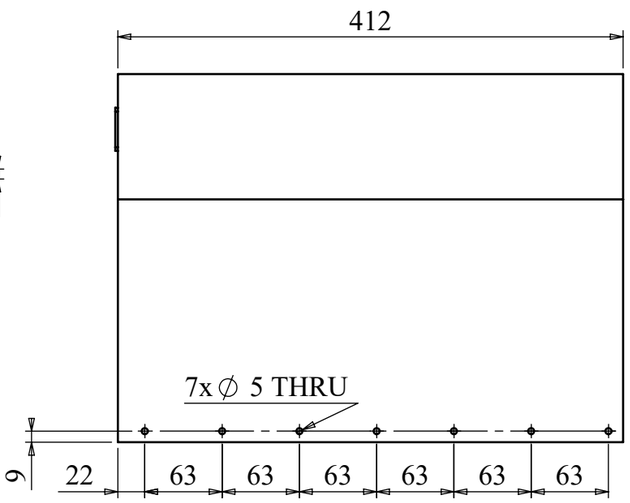


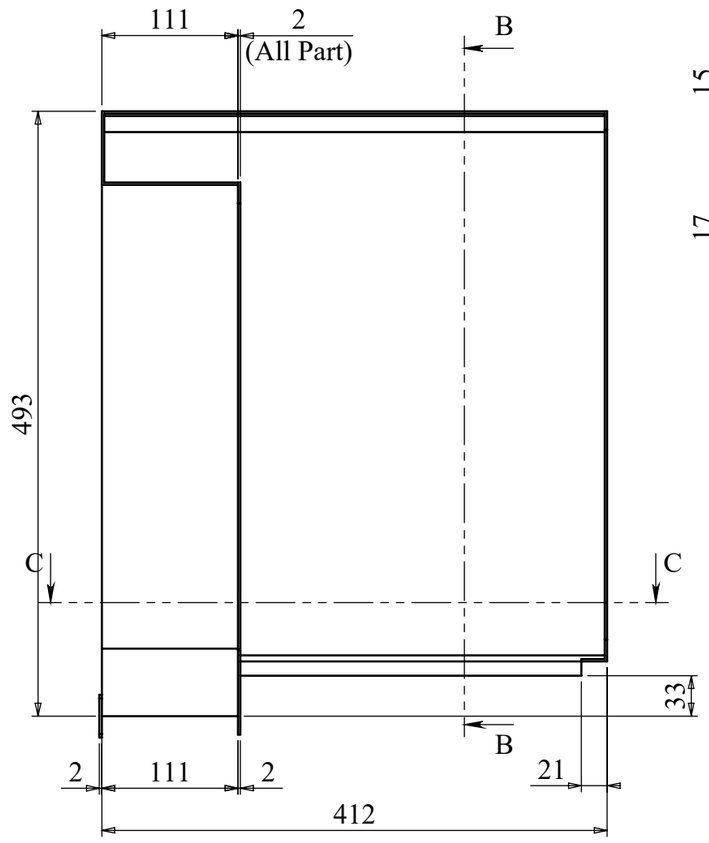
SECTION C-C

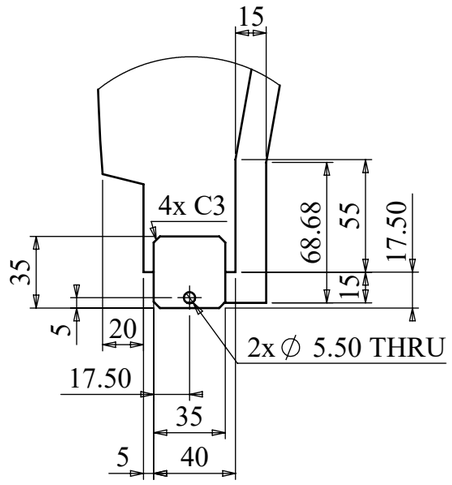


DETAIL U

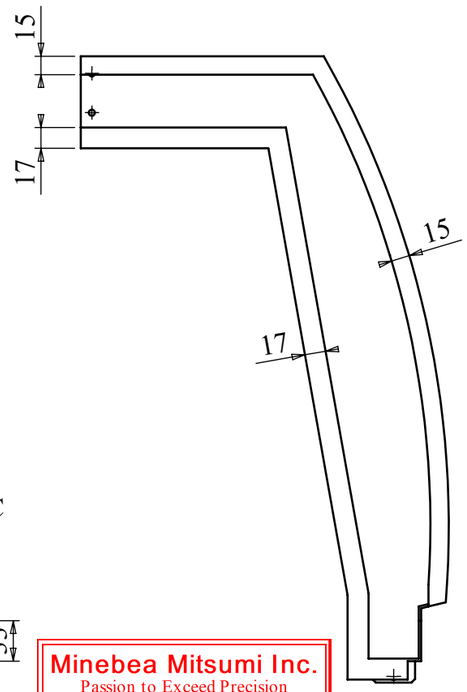


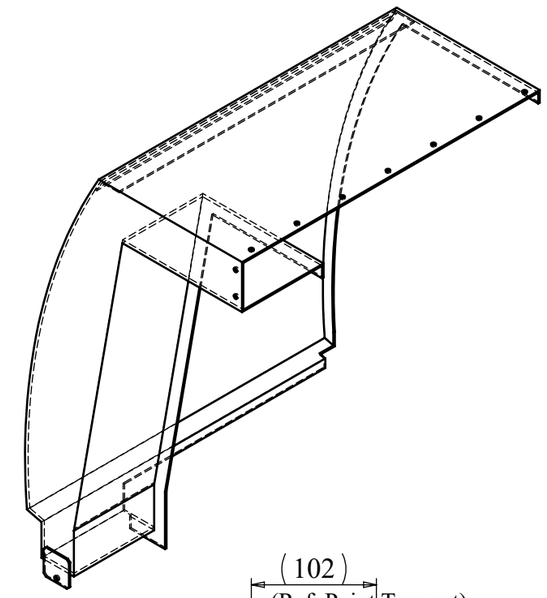


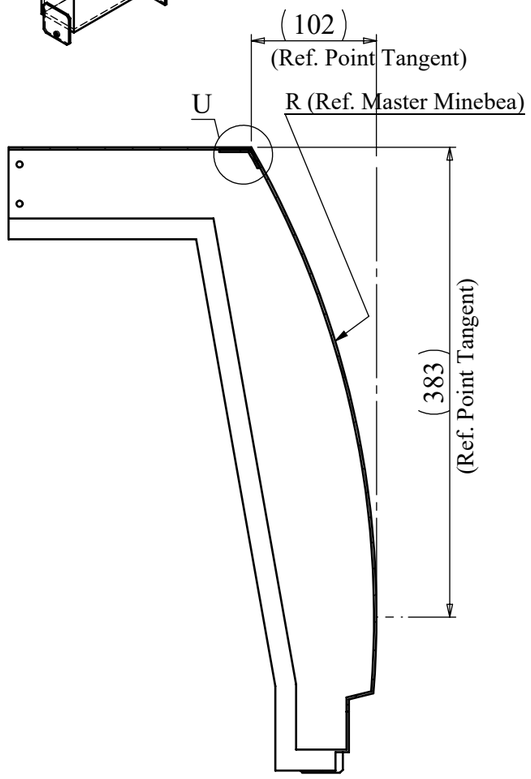




DETAIL V







SECTION B-B

Note

1. Unless otherwise specified all bending edges and corners is R1.0~R5.0
2. Accept welding
3. Paint the primer before the actual color.
4. Rework screw after actual color.
5. Baked Color ***(Ref.) Plate Master : DV-Minebea)

Right Cover

Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED C0.2 MAX or R0.2 MAX
GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C: Auto Loader Transfer [TSUGAMI-BW12]
MATERIAL: SPCC
FINISH: Powder Coating
HEAT TREATMENT: -

REMARK

3RD ANGLE PROJECTION

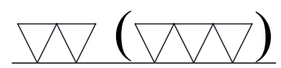
SCALE 1:5 | SHEET 3 / 4 | UNIT : mm

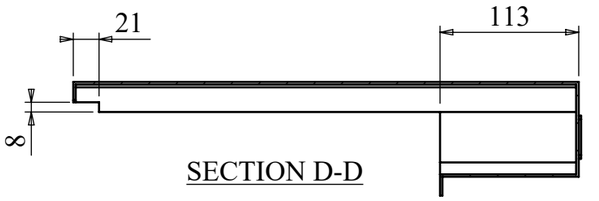
DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

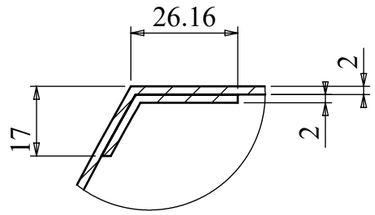
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

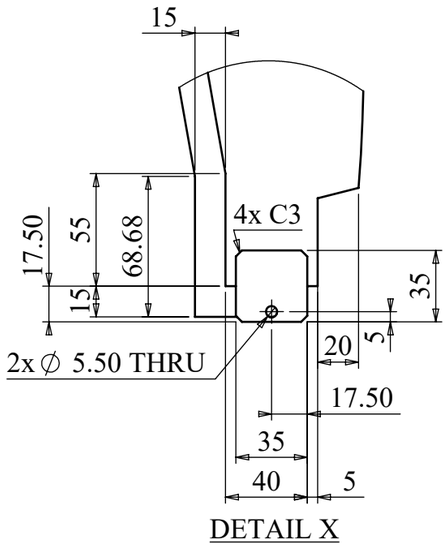
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

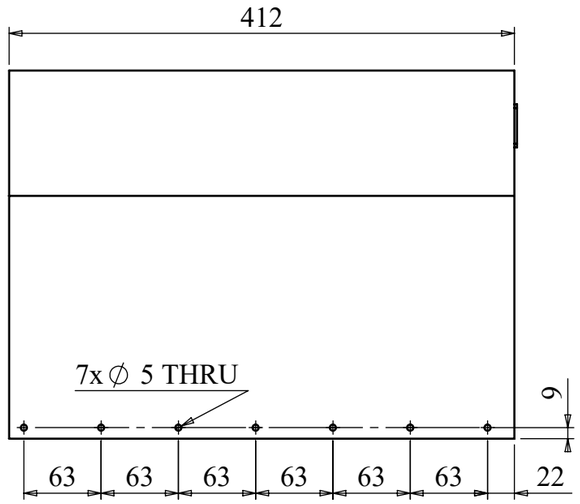
TITLE: Cover Door
DRAWING NUMBER: DVGM-ALW-MC-0010
1 REV.

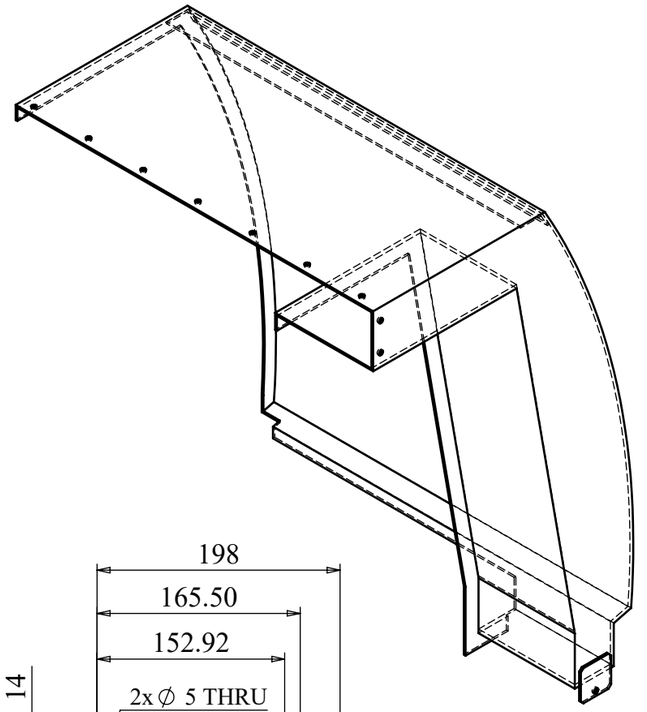


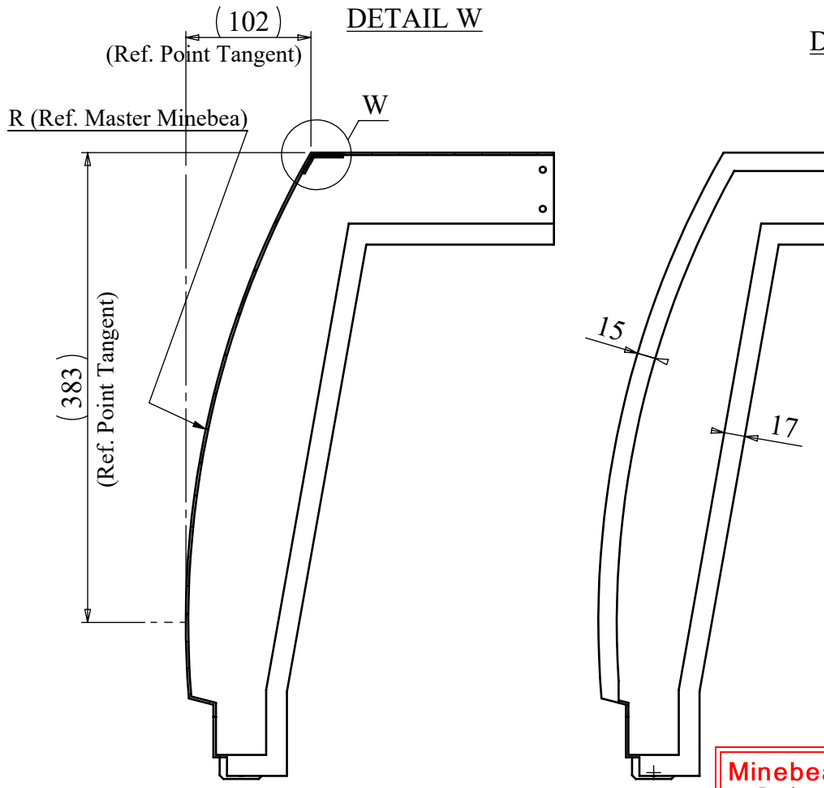


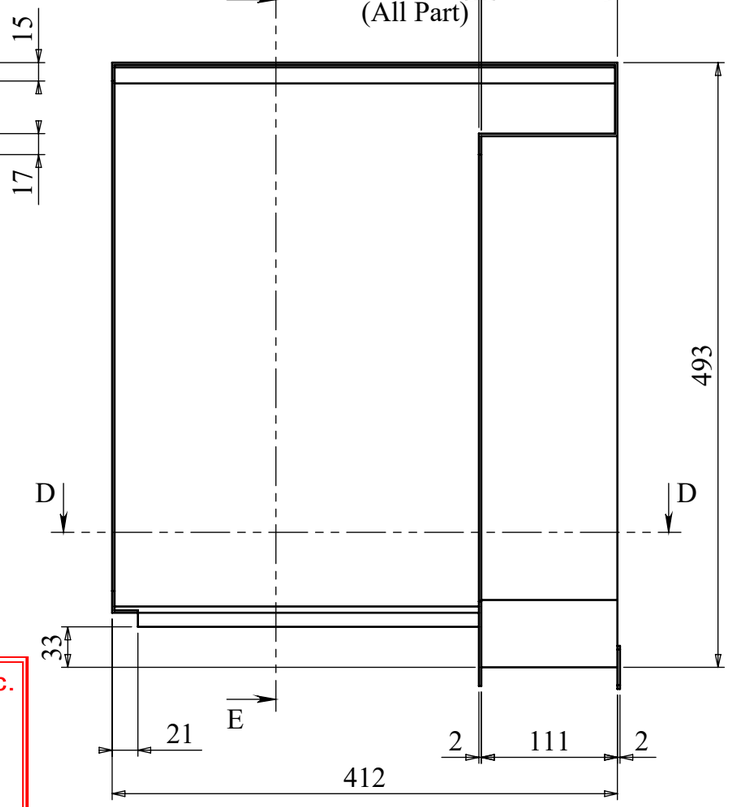


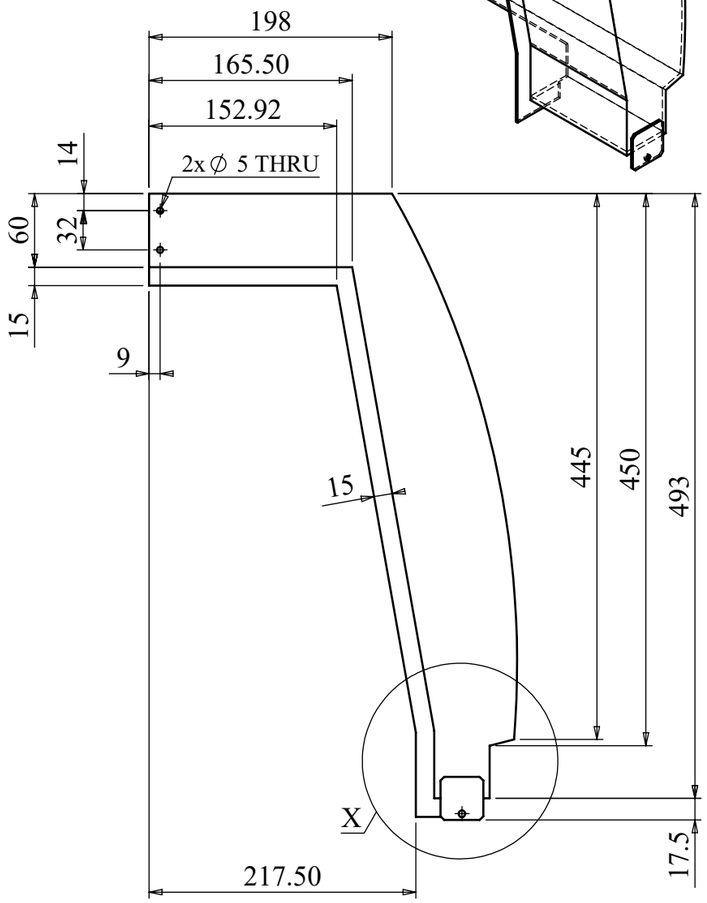












Note

1. Unless otherwise specified all bending edges and corners is R1.0~R5.0
2. Accept welding
3. Paint the primer before the actual color.
4. Rework screw after actual color.
5. Baked Color ***(Ref.) Plate Master : DV-Minebea)

Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Lift Cover

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED C0.2 MAX or R0.2 MAX	
GENERAL TOLERANCE	
JIS B 0405 MACHINING WORKING	
LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C: Auto Loader Transfer [TSUGAMI-BW12]
MATERIAL: SPCC
FINISH: Powder Coating
HEAT TREATMENT: -

REMARK

3RD ANGLE PROJECTION

SCALE 1:5 | SHEET 4 / 4 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

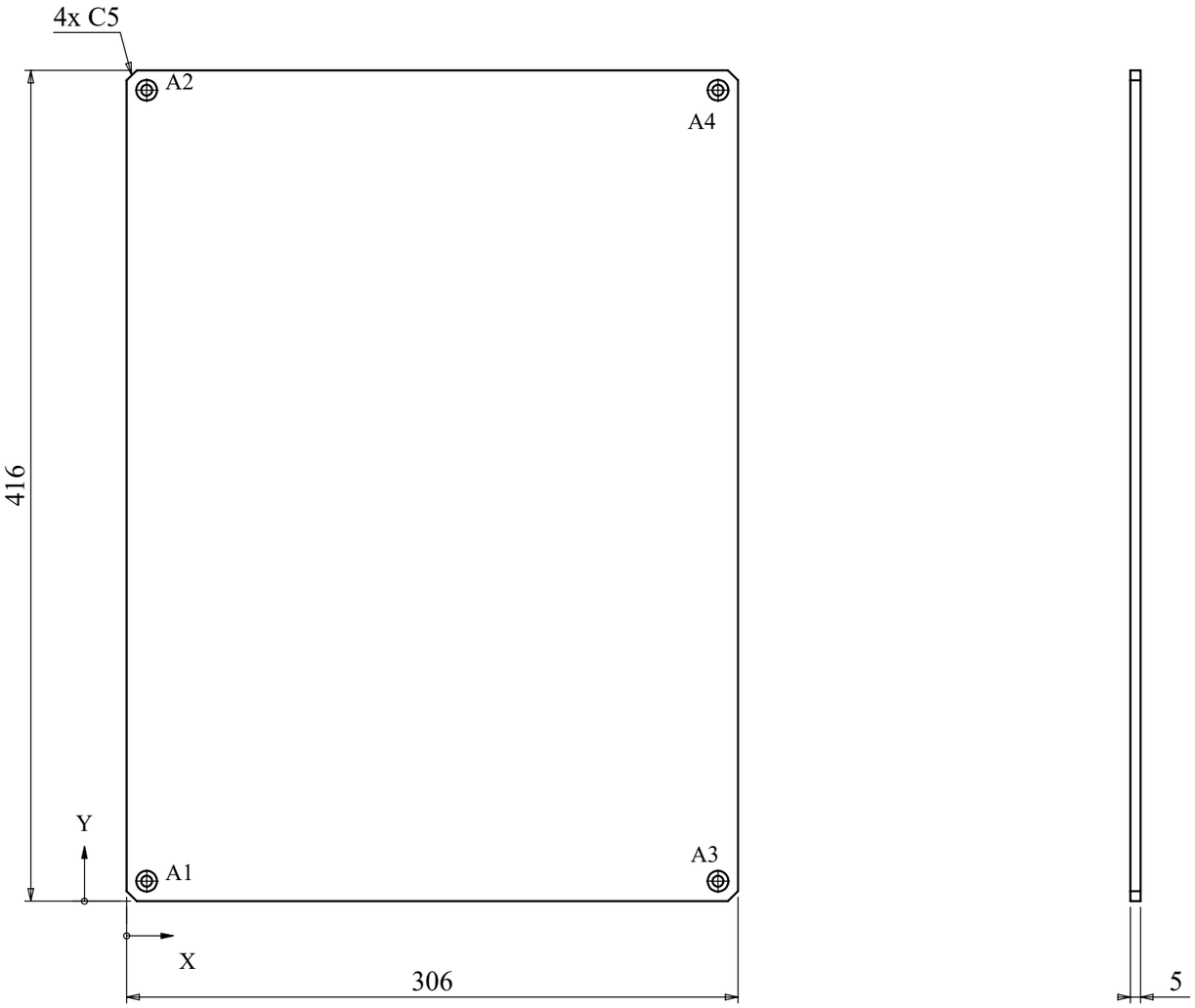
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE: Cover Door

DRAWING NUMBER: DVGM-ALW-MC-0010

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	10	10	∅ 5.50 THRU ALL ⌵ ∅ 10.40 X 90°
A2	10	406	
A3	296	10	
A4	296	406	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
Acrylic (Color NO.P-523)

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:3 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

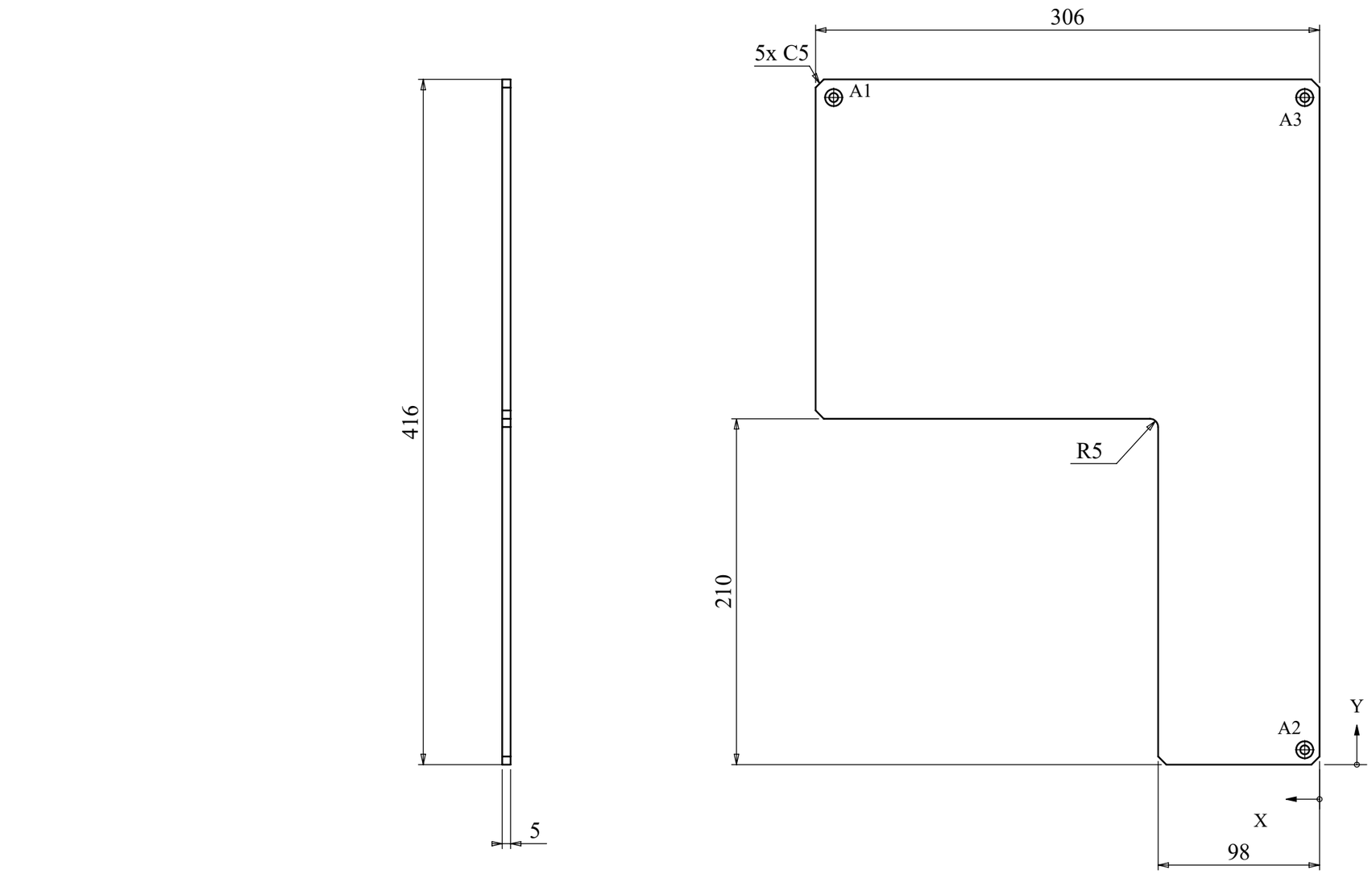
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE
Cover A

DRAWING NUMBER
DVGM-ALW-FS-0021

1 REV.



TAG	X LOC	Y LOC	SIZE
A1	295	405	∅ 5.50 THRU ALL ⌵ ∅ 10.40 X 90°
A2	9	9	
A3	9	405	

Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
Acrylic (Color NO.P-523)

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:3 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

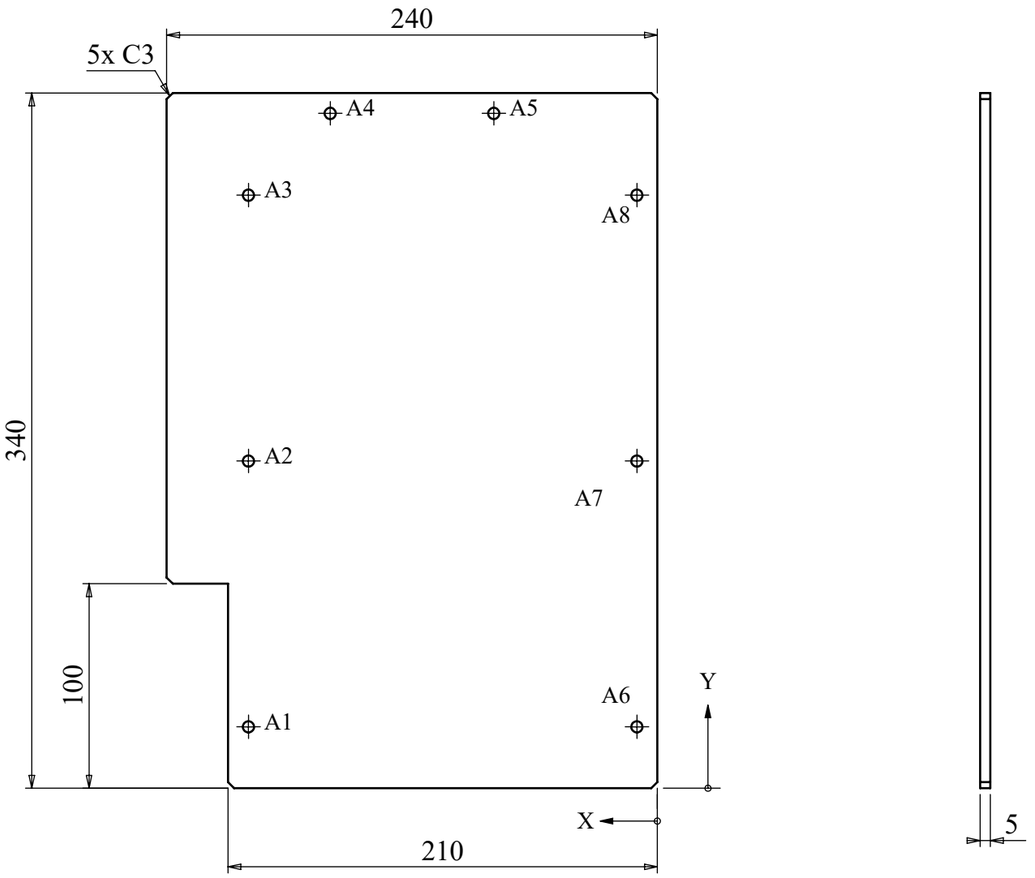
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

Cover B

DRAWING NUMBER
DVGM-ALW-FS-0031

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	200	30	∅ 5.50 THRU ALL
A2	200	160	
A3	200	290	
A4	160	330	
A5	80	330	
A6	10	30	
A7	10	160	
A8	10	290	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
Acrylic (Color NO.P-523)

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:3 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

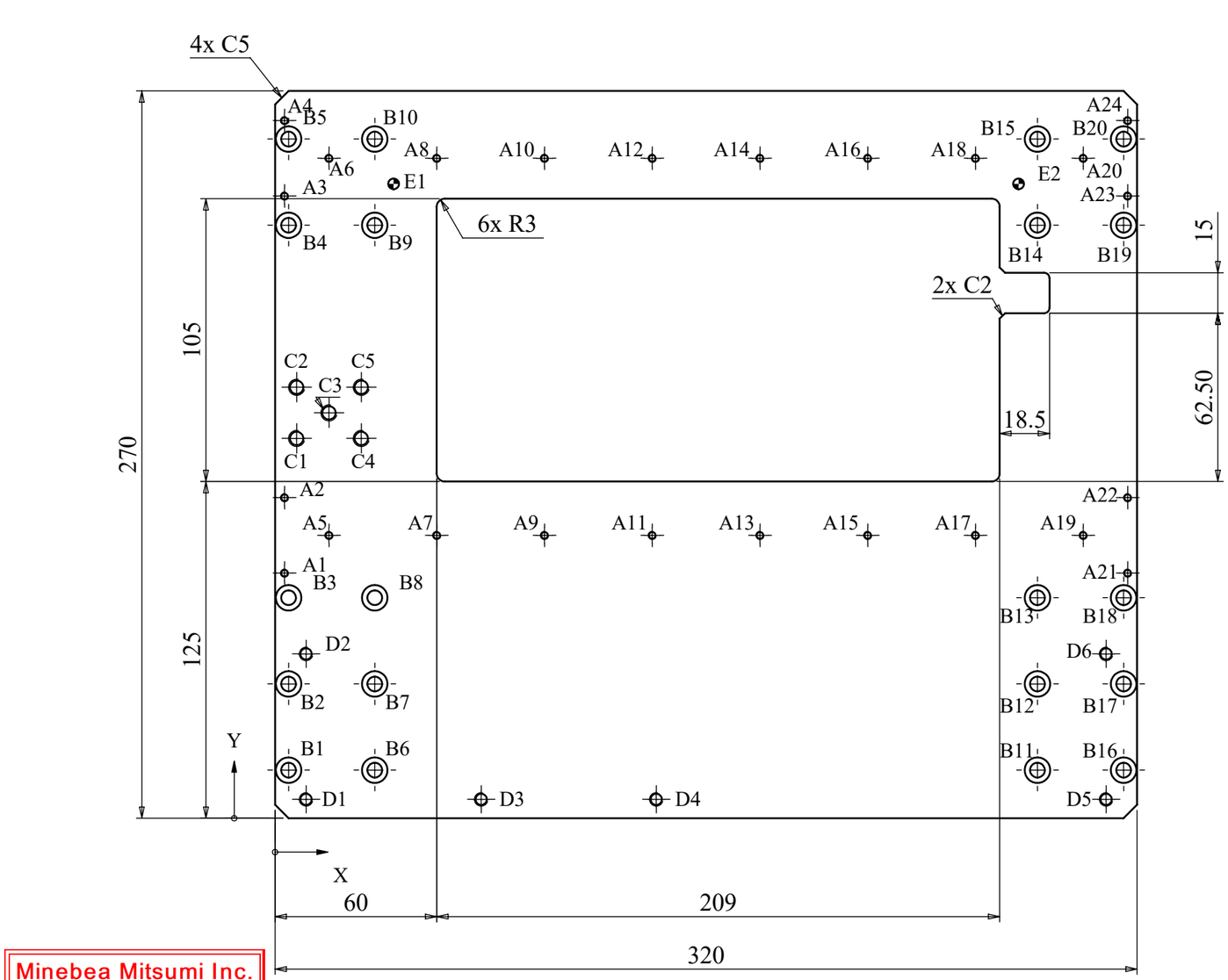
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

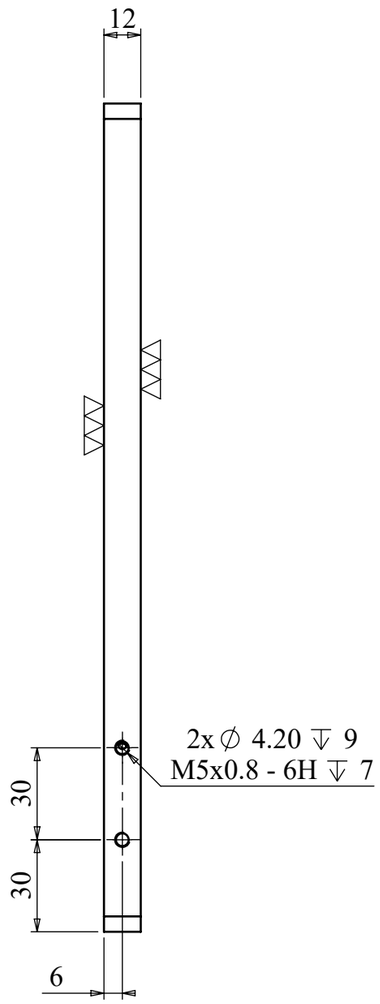
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

Partition

DRAWING NUMBER
DVGM-ALW-FS-0040

1 REV.





TAG	X LOC	Y LOC	SIZE
A1	3.50	91	∅ 2.50 THRU ALL M3x0.5 - 6H THRU ALL
A2	3.50	119	
A3	3.50	231	
A4	3.50	259	
A5	20	105	
A6	20	245	
A7	60	105	
A8	60	245	
A9	100	105	
A10	100	245	
A11	140	105	
A12	140	245	
A13	180	105	
A14	180	245	
A15	220	105	
A16	220	245	
A17	260	105	
A18	260	245	
A19	300	105	
A20	300	245	
A21	316.50	91	
A22	316.50	119	
A23	316.50	231	
A24	316.50	259	
B1	5	17.85	∅ 5.50 THRU ALL ⌴ ∅ 9.50 ↧ 5.50
B2	5	49.85	
B3	5	81.85	
B4	5	220.15	
B5	5	252.15	
B6	37	17.85	
B7	37	49.85	
B8	37	81.85	
B9	37	220.15	
B10	37	252.15	
B11	283	17.85	
B12	283	49.85	
B13	283	81.85	
B14	283	220.15	
B15	283	252.15	
B16	315	17.85	
B17	315	49.85	
B18	315	81.85	
B19	315	220.15	
B20	315	252.15	
C1	8	141	∅ 5 THRU ALL M6x1.0 - 6H THRU ALL
C2	8	160	
C3	20	150.50	
C4	32	141	
C5	32	160	

TAG	X LOC	Y LOC	SIZE
D1	11.50	7	∅ 4.20 THRU ALL M5x0.8 - 6H THRU ALL
D2	11.50	61	
D3	76.50	7	
D4	141.50	7	
D5	308.50	7	
D6	308.50	61	
E1	44	235.50	∅ 4 +0.01 0 THRU ALL
E2	276	235.50	

Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
A6063

FINISH
Clear Anodize

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:2 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

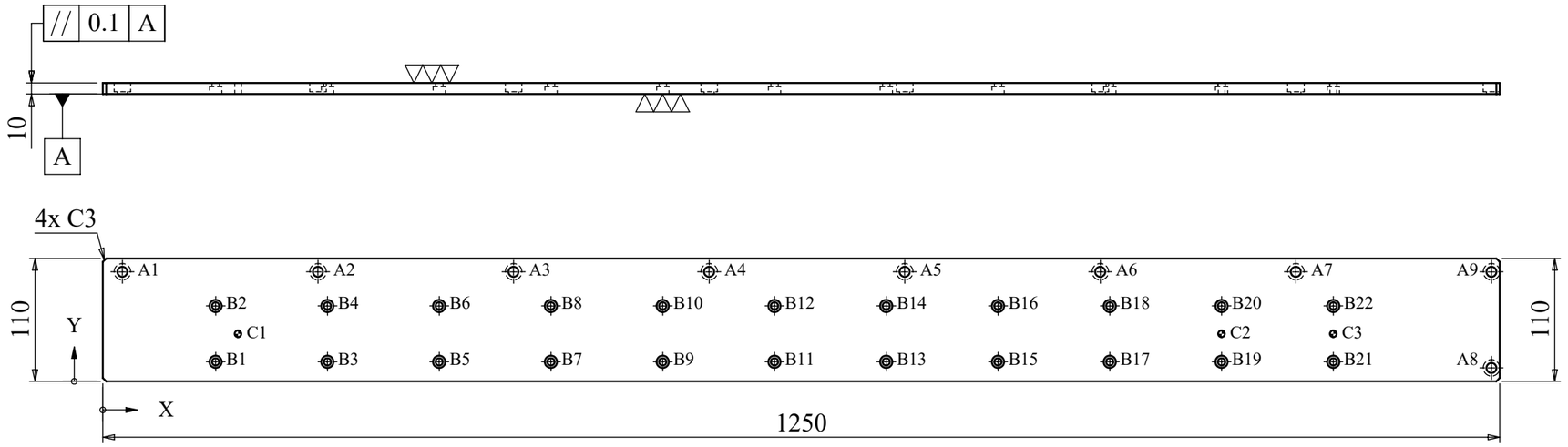
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

Plate Slide A

DRAWING NUMBER
DVGM-ALW-LU-0031

1 REV.



TAG	X LOC	Y LOC	SIZE
A1	17.50	98	⌀ 9 THRU ALL ⌴ ⌀ 14.80 ↧ 8 (Back Side)
A2	192.50	98	
A3	367.50	98	
A4	542.50	98	
A5	717.50	98	
A6	892.50	98	
A7	1067.50	98	
A8	1242.50	12	
A9	1242.50	98	
B1	101	17.50	⌀ 6.60 THRU ALL ⌴ ⌀ 11 ↧ 6.50
B2	101	67.50	
B3	201	17.50	
B4	201	67.50	
B5	301	17.50	
B6	301	67.50	
B7	401	17.50	
B8	401	67.50	
B9	501	17.50	
B10	501	67.50	
B11	601	17.50	
B12	601	67.50	
B13	701	17.50	
B14	701	67.50	
B15	801	17.50	
B16	801	67.50	
B17	901	17.50	
B18	901	67.50	
B19	1001	17.50	
B20	1001	67.50	
B21	1101	17.50	
B22	1101	67.50	
C1	121 ±0.02	42.50 ±0.02	+0.01 ⌀ 6 0 THRU ALL
C2	1001 ±0.02	42.50 ±0.02	
C3	1101 ±0.02	42.50 ±0.02	

Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
A6063

FINISH
Clear Anodize

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:5 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

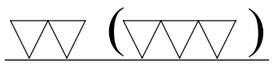
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

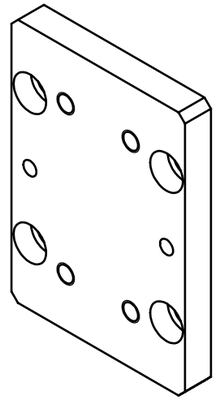
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

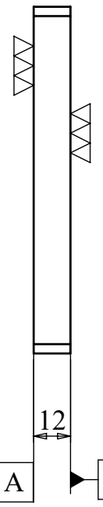
Plate Robo

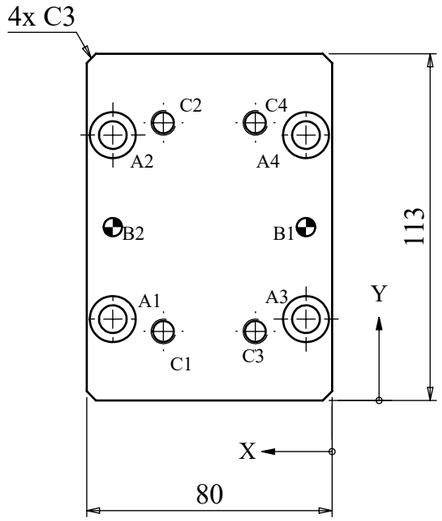
DRAWING NUMBER
DVGM-ALW-LU-0281

1 REV.









Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	71.50	26.50	⌀ 9 THRU ALL ⌴ ⌀ 15 ↧ 8
A2	71.50	86.50	
A3	8.50	26.50	
A4	8.50	86.50	
B1	8.50 ±0.02	56.50 ±0.02	+0.01 ⌀ 6 0 THRU ALL
B2	71.50 ±0.02	56.50 ±0.02	
C1	55	22.50	⌀ 6.80 THRU ALL M8x1.25 - 6H THRU ALL
C2	55	90.50	
C3	25	22.50	
C4	25	90.50	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	-- -- --	-- -- --	-- -- --	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
SS400

FINISH
Nickel Plating

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:2 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

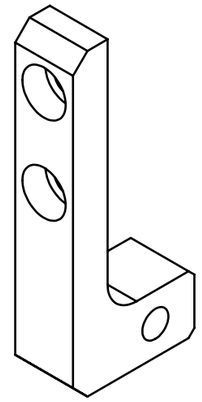
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

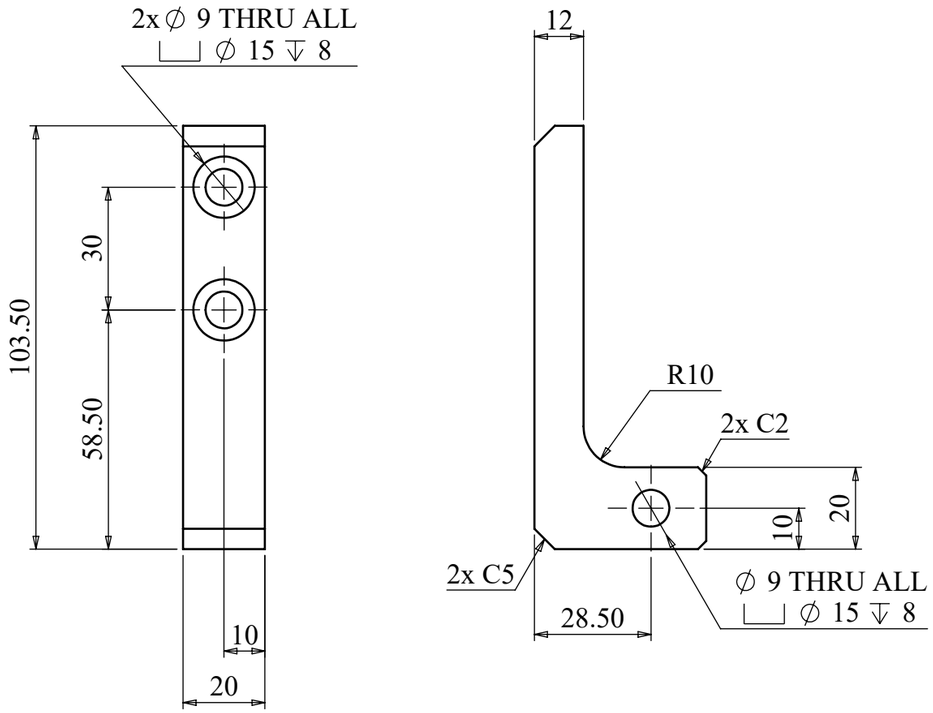
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE
Slide Robo

DRAWING NUMBER
DVGM-ALW-LU-0291

REV. 1





Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
SS400

FINISH
Nickel Plating

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:1.5 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

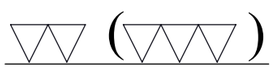
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

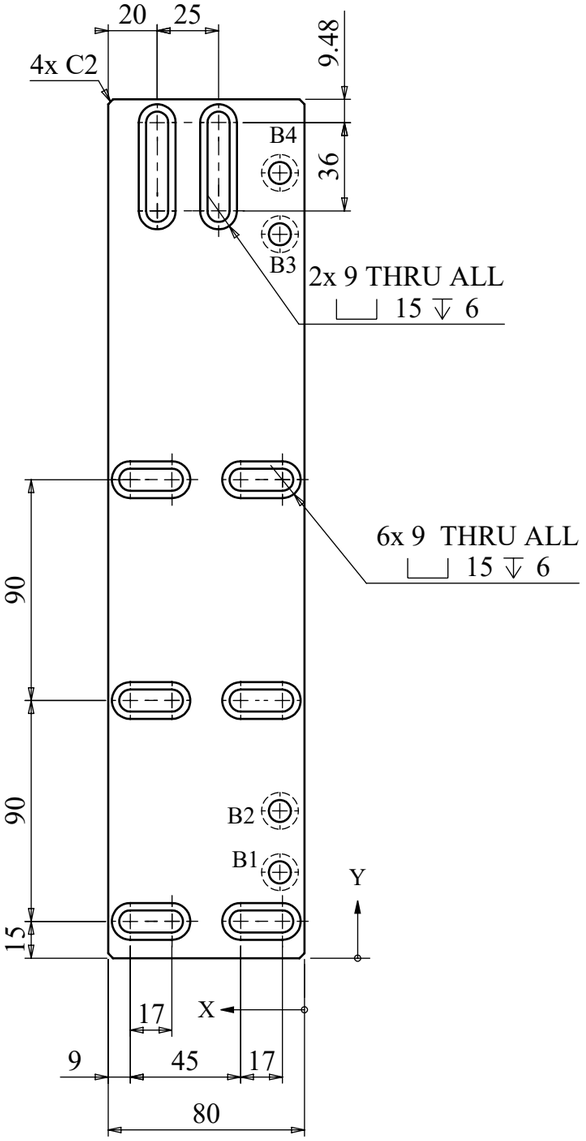
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

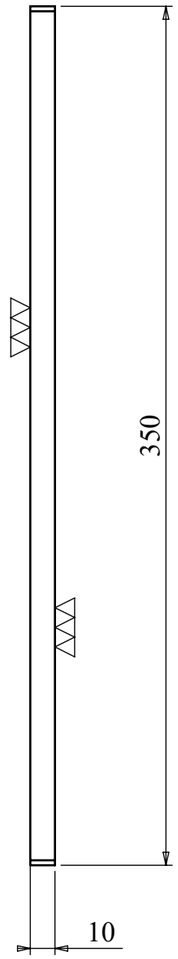
Plate Pusher

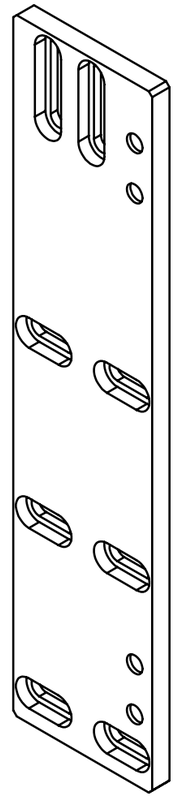
DRAWING NUMBER
DVGM-ALW-LU-0301

1 REV.









Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
B1	10	35	∅ 9 THRU ALL ⌴ ∅ 15 ↧ 8 (Back Side)
B2	10	60	
B3	10	295	
B4	10	320	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
SS400

FINISH
Nickel Plating

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:2.5 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

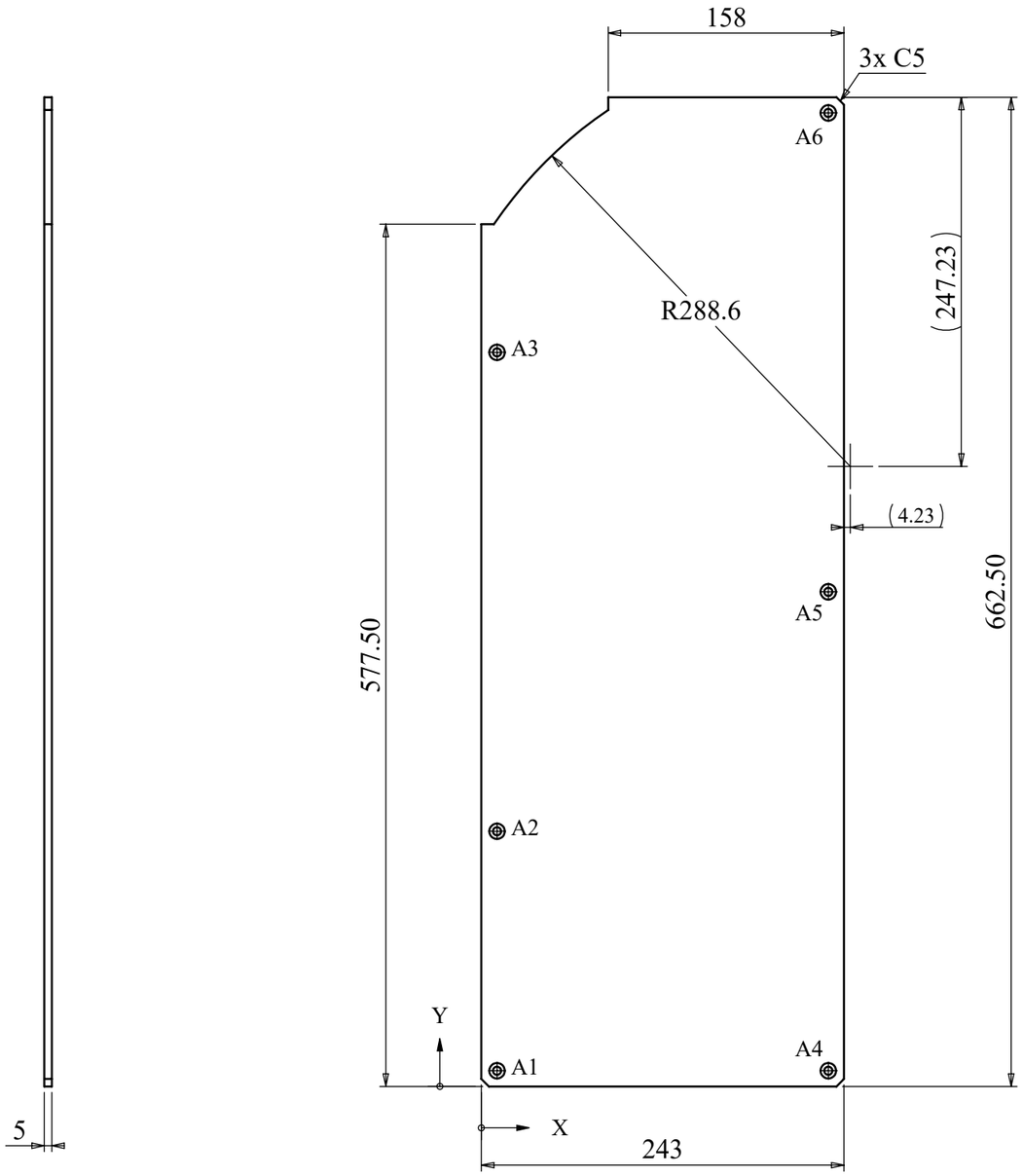
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

Bottom Plate

DRAWING NUMBER
DVGM-ALW-LU-0341

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	10.50	10.50	∅ 5.50 THRU ALL ⌵ ∅ 10.40 X 90°
A2	10.50	170.88	
A3	10.50	491.63	
A4	232.50	10.50	
A5	232.50	331.25	
A6	232.50	652	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
Acrylic (Color NO.P-523)

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:4 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

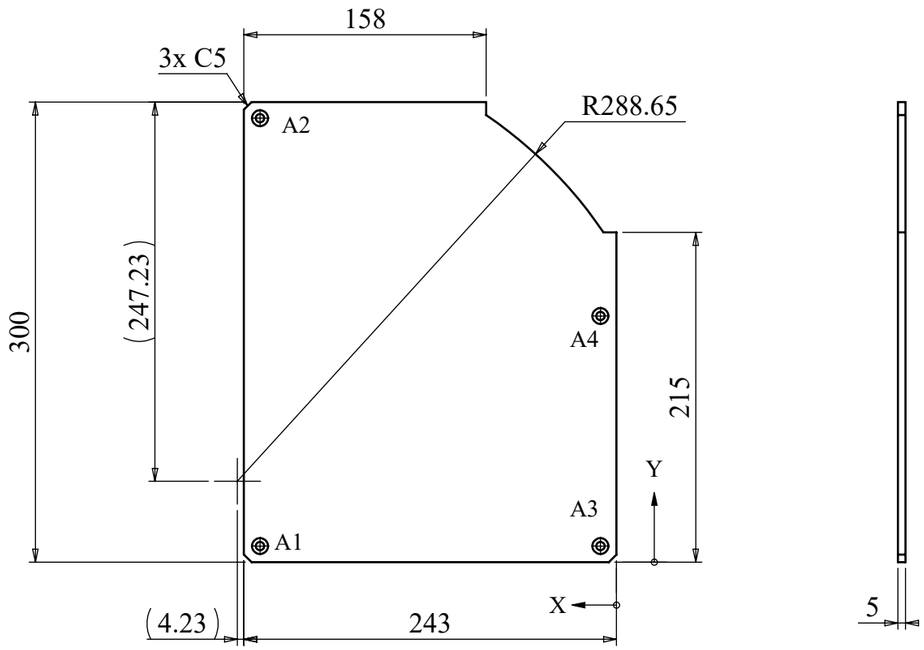
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

Cover AA

DRAWING NUMBER
DVGM-ALW-LU-0361

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	232.50	10.50	∅ 5.50 THRU ALL ◡ ∅ 10.40 X 90°
A2	232.50	289.50	
A3	10.50	10.50	
A4	10.50	160.50	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
Acrylic (Color NO.P-523)

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:4 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

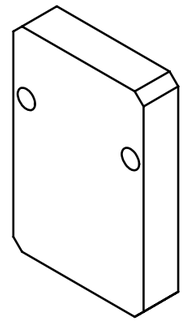
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

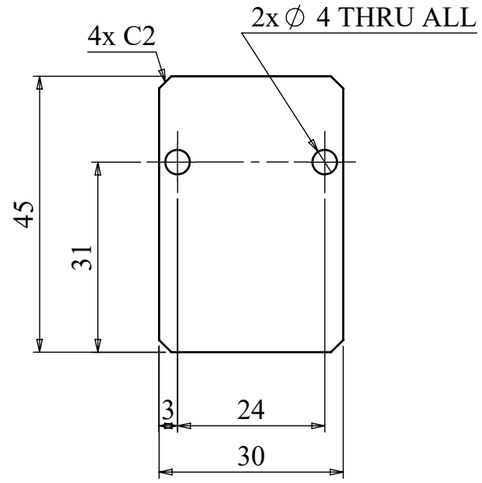
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

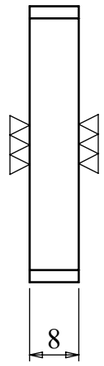
Cover AC

DRAWING NUMBER
DVGM-ALW-LU-0381

1 REV.







Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	-- -- --	-- -- --	-- -- --	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
A6063

FINISH
Clear Anodize

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:1 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

DATE : 24. Apr. 2025 | Qt'y : - Pcs.

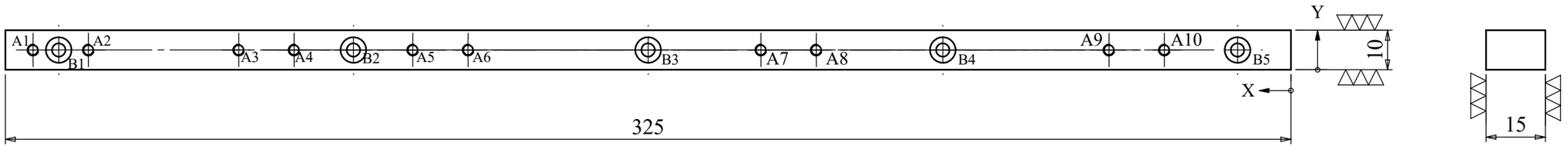
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

DWG

Plate Support

DRAWING NUMBER
DVGM-ALW-LU-0390

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	318	5	∅ 2.50 THRU ALL M3x0.5 - 6H THRU ALL
A2	304	5	
A3	266	5	
A4	252	5	
A5	222	5	
A6	208	5	
A7	134	5	
A8	120	5	
A9	46	5	
A10	32	5	
B1	311.50	5	∅ 3.40 THRU ALL ⌴ ∅ 6.50 ↧ 3.50
B2	237	5	
B3	162.50	5	
B4	88	5	
B5	13.50	5	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	-- -- --	-- -- --	-- -- --	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
PVC GRAY

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:1.2 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

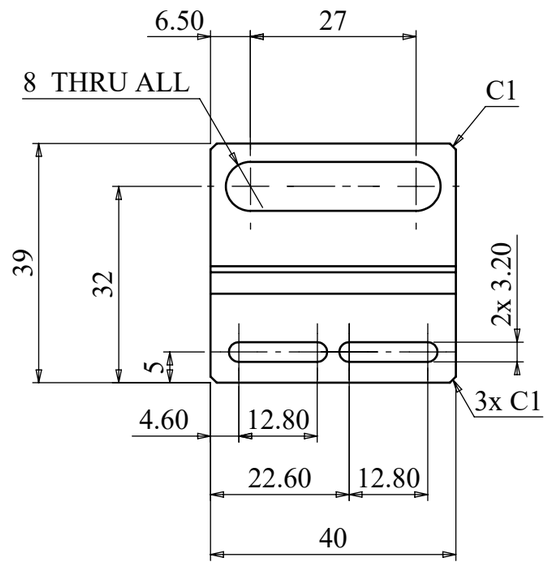
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

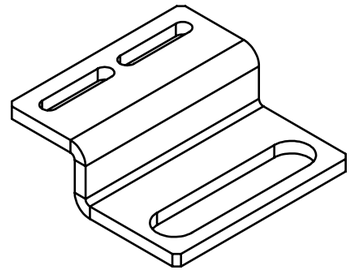
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

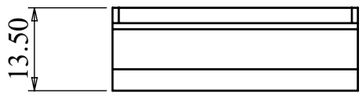
TITLE
Support Side

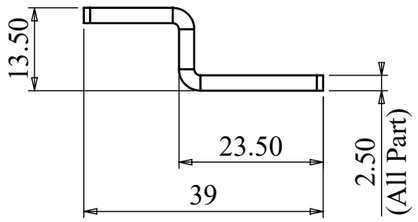
DRAWING NUMBER
DVGM-ALW-ST-0041

1 REV.









Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Note

1. Unless otherwise specified all bending edges and corners is R1.0~R5.0

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
SUS304

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:1 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

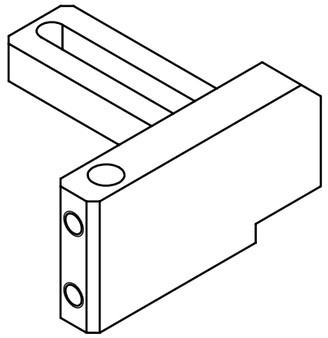
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

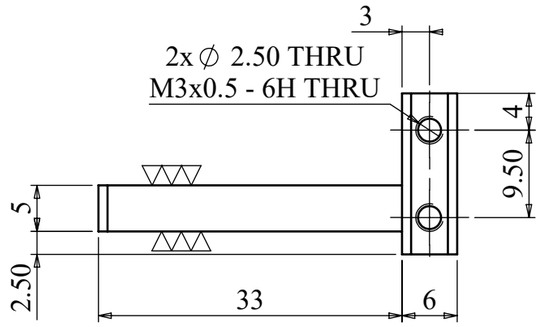
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

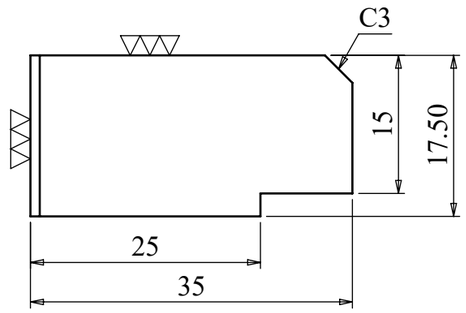
Stand Sensor

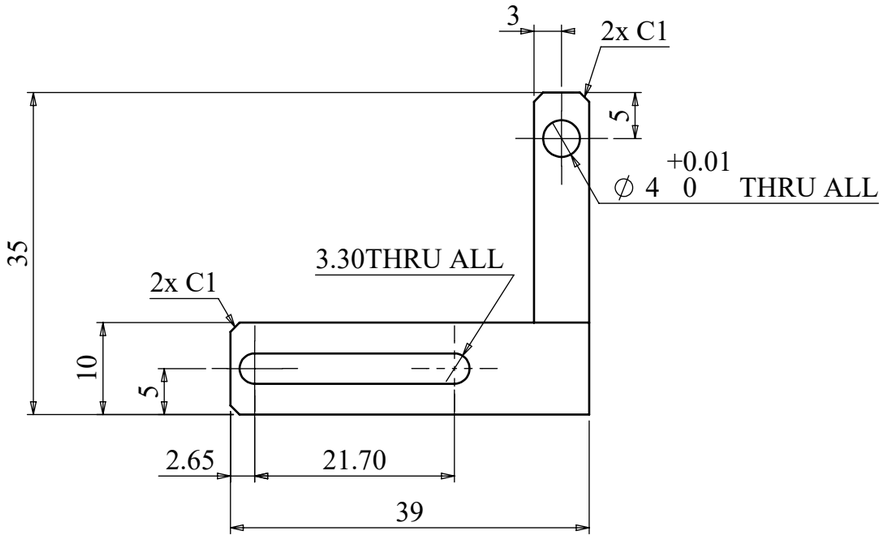
DRAWING NUMBER
DVGM-ALW-ST-0081

1 REV.









Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
SUS304

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1.5:1 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

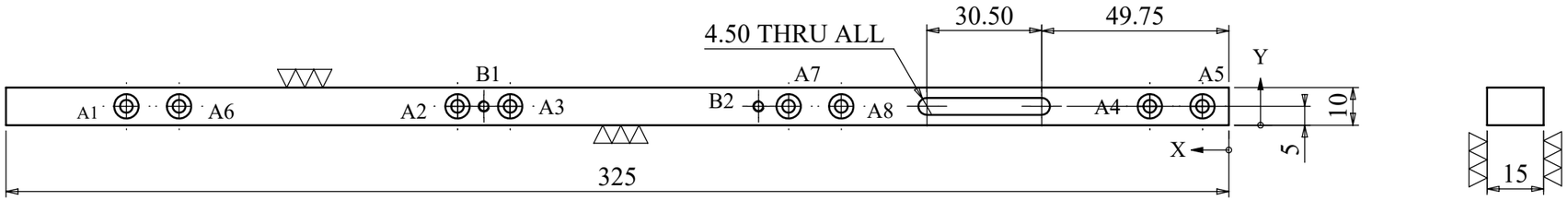
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

Plate Sensor

DRAWING NUMBER
DVGM-ALW-ST-0280

1 REV.



TAG	X LOC	Y LOC	SIZE
E1	259.50	9.20	∅ 2.50 THRU ALL M3x0.5 - 6H THRU ALL
E2	232.50	9.20	
E3	89.50	9.20	
E4	62.50	9.20	
E5	37.50	9.20	
E6	10.50	9.20	

TAG	X LOC	Y LOC	SIZE
A1	293	5	∅ 3.40 THRU ALL ⌴ ∅ 6.50 ↧ 3.50
A2	205	5	
A3	191	5	
A4	21	5	
A5	7	5	
A6	279	5	
A7	117	5	
A8	103	5	
B1	198	5	∅ 2.50 THRU ALL M3x0.5 - 6H THRU ALL
B2	125	5	

Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
PVC GRAY

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:1.5 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

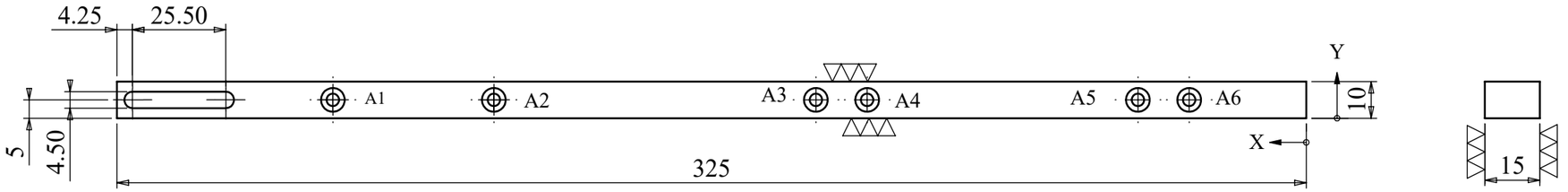
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE
Plate Support F

DRAWING NUMBER
DVGM-ALW-ST-0290

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE	TAG	X LOC	Y LOC	SIZE
E1	279.50	9.20	∅ 2.50 THRU ALL M3x0.5 - 6H THRU ALL	A1	266	5	∅ 3.40 THRU ALL ⌴ ∅ 6.50 ↧ 3.50
E2	252.50	9.20		A2	222	5	
E3	150.50	9.20		A3	134	5	
E4	123.50	9.20		A4	120	5	
E5	101.50	9.20		A5	46	5	
E6	74.50	9.20		A6	32	5	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
PVC GRAY

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:1.5 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

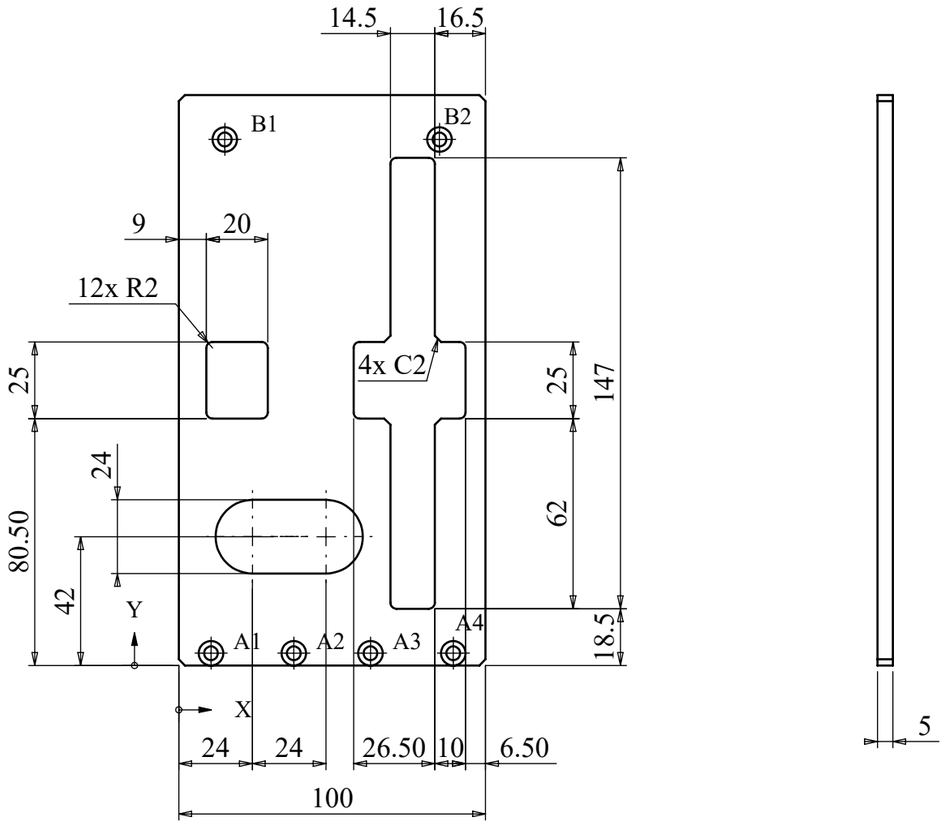
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE
Plate Support B

DRAWING NUMBER
DVGM-ALW-ST-0300

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	10.50	4	∅ 4.50 THRU ALL ⌵ ∅ 8 X 90°
A2	37.50	4	
A3	62.50	4	
A4	89.50	4	
B1	15	171.50	∅ 4.50 THRU ALL ⌵ ∅ 8 X 90°
B2	85	171.50	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C: Auto Loader Transfer [TSUGAMI-BW12]
MATERIAL: Acrylic (Color NO.P-523)
FINISH: -
HEAT TREATMENT: -

REMARK

3RD ANGLE PROJECTION

SCALE 1:2 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

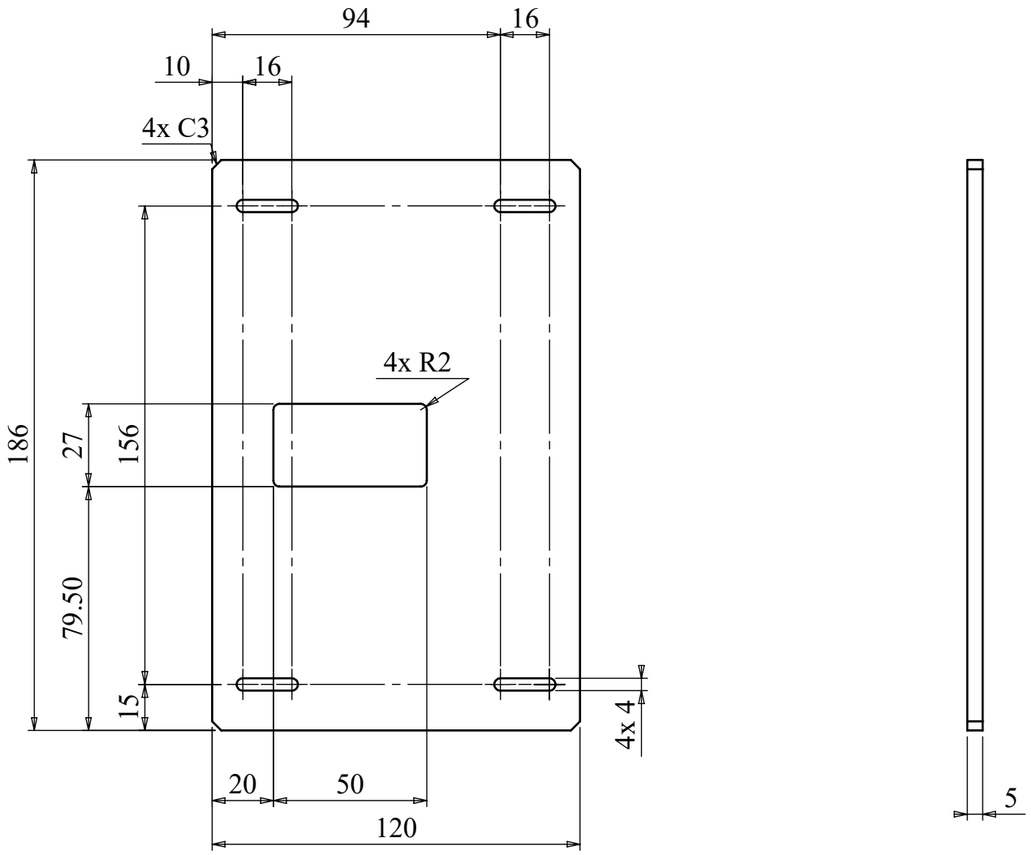
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE: Cover Tray A

DRAWING NUMBER: DVGM-ALW-ST-0310

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S, 6.3S, 25S, 100S, UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
Acrylic (Color NO.P-523)

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:2 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

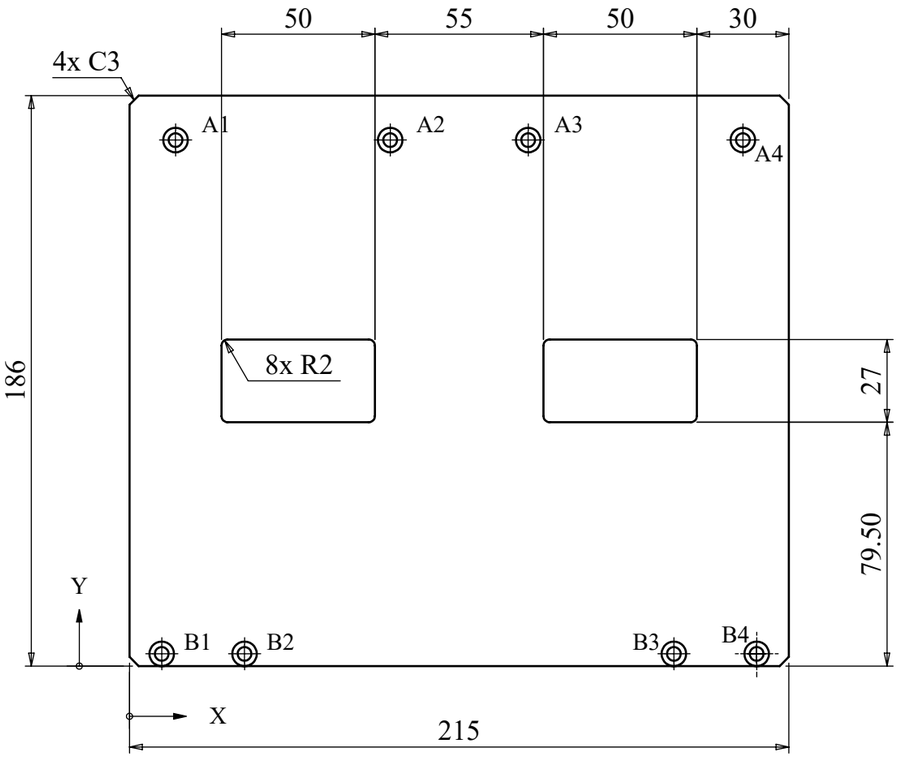
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

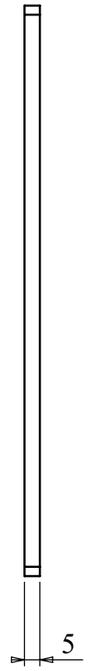
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE
Cover Tray B

DRAWING NUMBER
DVGM-ALW-ST-0320

1 REV.





Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	15	171.50	∅ 4.50 THRU ALL ⌵ ∅ 8 X 90°
A2	85	171.50	
A3	130	171.50	
A4	200	171.50	
B1	10.50	4	∅ 4.50 THRU ALL ⌵ ∅ 8 X 90°
B2	37.50	4	
B3	177.50	4	
B4	204.50	4	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S ▽ 6.3S ▽ 25S ▽ 100S ▽ UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
Acrylic (Color NO.P-523)

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:2 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

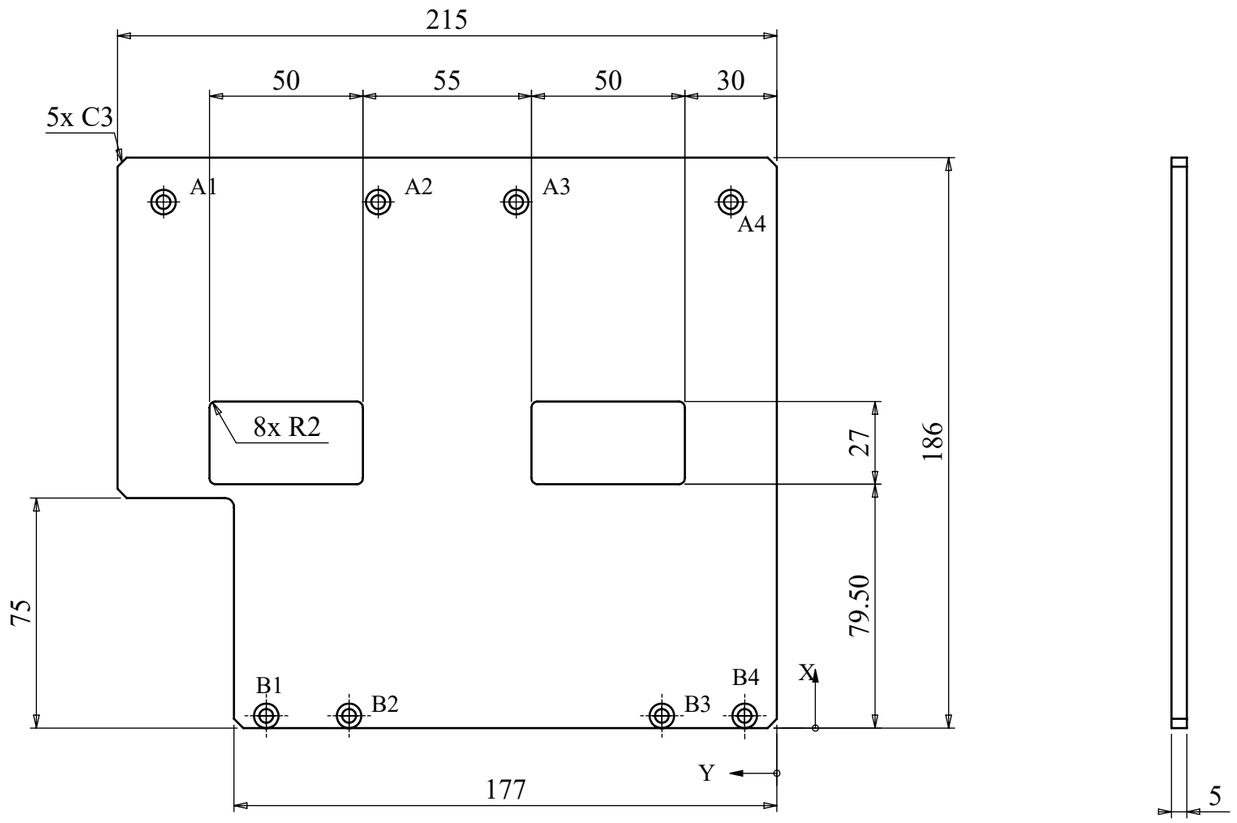
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE
Cover Tray C

DRAWING NUMBER
DVGM-ALW-ST-0330

1 REV.



Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

TAG	X LOC	Y LOC	SIZE
A1	171.50	200	∅ 4.50 THRU ALL ⌵ ∅ 8 X 90°
A2	171.50	130	
A3	171.50	85	
A4	171.50	15	
B1	4	166.50	∅ 4.50 THRU ALL ⌵ ∅ 8 X 90°
B2	4	139.50	
B3	4	37.50	
B4	4	10.50	

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405 MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S 6.3S 25S 100S UNCUT

M / C: Auto Loader Transfer [TSUGAMI-BW12]
MATERIAL: Acrylic (Color NO.P-523)
FINISH: -
HEAT TREATMENT: -

REMARK

3RD ANGLE PROJECTION

SCALE 1:2 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

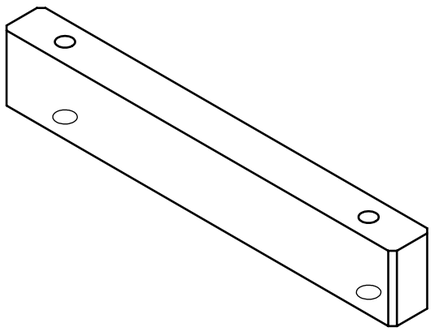
DATE : 24. Apr. 2025 | Qt'y : - Pcs.

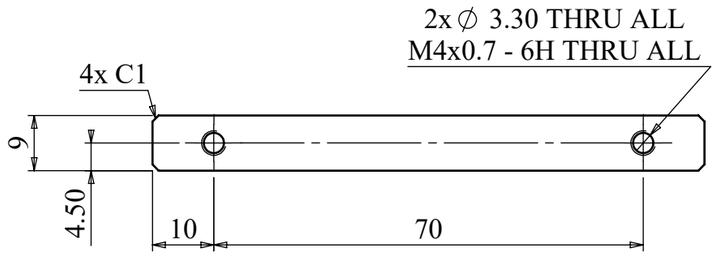
NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

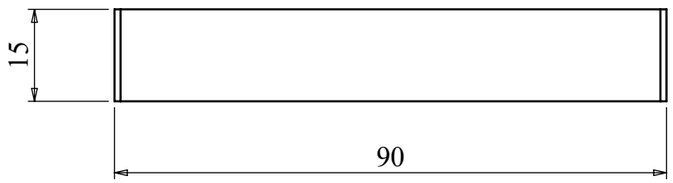
TITLE: Cover Tray D

DRAWING NUMBER: DVGM-ALW-ST-0340

1 REV.







Minebea Mitsumi Inc.
Passion to Exceed Precision
Uncontrolled Copy
25. Apr. 25
ISSUED
G/M 1st Div. Machine Development

Rev.	ECN No.	DATE	PERSON	DESCRIPTION
1	------	------	------	DEVELOPMENT RELEASE

UNLESS SPECIFIED
C0.2 MAX or R0.2 MAX

GENERAL TOLERANCE
JIS B 0405
MACHINING WORKING

LESS THAN 0.5	± 0.05
0.5 ~ 6	± 0.1
6 ~ 30	± 0.2
30 ~ 120	± 0.3
120 ~ 400	± 0.4
400 ~ 1000	± 0.8
ANGLES	± 1°

0.8S / 6.3S / 25S / 100S / UNCUT

M / C
Auto Loader Transfer [TSUGAMI-BW12]

MATERIAL
PVC GRAY

FINISH
-

HEAT TREATMENT
-

REMARK

3RD ANGLE PROJECTION

SCALE 1:1 | SHEET 1 / 1 | UNIT : mm

DETAILED	DESIGNED	CHECKED	APPROVED
P.Tawan	P.Naiwinit	P.Naiwinit	B.Komgrich

DATE : 24. Apr. 2025 | Qt'y : - Pcs.

NMB MinebeaMitsumi Inc.
Machine Development Section General Machining Parts 1st Division

TITLE
Plate Bottom

DRAWING NUMBER
DVGM-ALW-ST-0350

1 REV.